Privacy attacks and differential privacy



AI security · University · Lecture 3

In this lecture

- Explain membership inference.
- Define differential privacy.
- Calibrate noise.

Remember

- Models leak behaviour.
- Scale = Δ/ε.
- Budget the privacy.

Respond at random

1. Warm-up (5 min): Is it anonymous?
2. Flip (10 min): Respond.
3. Estimate (15 min): Recover.
4. Calibrate (25 min): Noise.
5. Wrap-up (5 min): One-minute paper.

Let’s discuss

1. Why can no one be blamed for an answer?
2. What does ε cost?
3. Who should set the budget?

Exit ticket

- Which data release in your institution would you protect with differential privacy?